

Coding & Solution

Date	05 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Solution Summary

Repository Link / URL	[Add your GitHub repository link here]
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Bootstrap 5
Key Features Implemented	Loan eligibility prediction, data preprocessing pipeline, model comparison, confidence & risk scoring, responsive web UI
Pending / Incomplete Features	User authentication, database integration, explainable AI (SHAP/LIME)
Setup / Run Instructions	pip install -r requirements.txt → python app.py → open http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / routes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes: The application is modular, easy to maintain, and uses a trained K-Nearest Neighbors model (84.55% testing accuracy) for loan eligibility prediction.

Application Screenshots

Home Page – Loan Application Form

 **Smart Lender**
AI Powered Loan Eligibility Prediction System

Gender

Male

Dependents

0

Self Employed

No

Applicant Income

Loan Amount

Credit History

Good Credit History

Married

Yes

Education

Graduate

Property Area

Urban

Co Applicant Income

Loan Term

360

Predict Loan Eligibility

4
ML Models

KNN
Best Model

84.55%
Accuracy

AI
Prediction